AT - SUB Click/Tan MNO hosted

Country	Austria
Service Type	Subscription
Flow	CLICK, WEB TAN
Hosting:	Pages by MNO, TAN by MNO
Functionality	Standard; Immediate Subscription

1. Flow Description

This Flow allows a Subscription Service in Austria.

A MERCHANT provides e.g. a xyz service, or sells zyx and an INTEGRATIOR acts between DIMOCO and MERCHANT (MERCHANT/INTEGRATOR can be the same instance).

The MNO hosts the payment pages and handles the TAN message and verification. Payment is fulfilled at once and content delivery happens **after** the successful payment.

2. Terminology

- USER … consumer (enduser) buying a merchant's product
- INTEGRATOR … technical Integrator providing the technical communication between MERCHANT, DIMOCO and USER. This can be also the MERCHANT.
- MERCHANT … service provider offering, the service the USER pays for (e.g. Product, Access…)
- MNO … mobile network operator
- MSISDN … mobile subscriber integrated services digital network number
- TransactionID … long-lived unique identifier for each USER transaction (generated by DIMOCO)
- SubscriptionID ... longlived unique identifier for each enduser subscription (generated by DIMOCO)
- pay:smart … DIMOCO's billing API
- MO-SMS ... SMS triggered by the USER (mobile originated)
- MT-SMS ... SMS sent to the USER (mobile terminated)
- CLICK Flow
    - USER is browsing on mobile device via mobile web (e.g. 3G/4G/5G)
    - Confirmation of purchase via click the confirmation button on the payment confirmation page
- WEB TAN Flow
    - USER is browsing on Wi-Fi or Desktop

    - Entering MSISDN in browser on the according MSISDN Entry Page
    - Confirmation via entering TAN on the TAN Entry Page

3. Credentials

- Merchant: 200416
- Password (secret-key): [redacted]
- Order: 115511

4. Settings

- Payment API: pay:SMART
- Service type: Subscription
- Tariff: 4,90 EUR
- Periodicity: 1 per week
- pay:periodic:
    - ENABLED - renewals triggered by DIMOCO
    - (Option amendable! Read below in point **Server2server communication** and **Renewal**)
- Service number: 0800 100 336
- Initial service page:
- Flow:
    - mobile web: CLICK (this is the default case, the user is on his mobile connection)
    - Wifi: WEB TAN (**only in case the user is not on his mobile connection**)
- Pages hosted by:
    - Click payment page: MNO
    - MSISDN entry page: INTEGRATOR
    - TAN Entry Page: MNO
    - Outcome page/Content Page: INTEGRATOR
- Messages hosted by:
    - TAN MT: MNO
    - Welcome MT: MNO
    - FAGG MT: DIMOCO
    - Stop confirmation MT: DIMOCO

5. Server2server communication

- Close notification URL: For API notification “close-subscription” in XML format
    - Either Share a static endpoint,
    - Or add one during API action “start-subscription” in parameter close_notification_url_callback.
- Renewal notification URL (renewal handling by DIMOCO): For API notification “renew-subscription” in XML format.

    - Either Share a static endpoint,
    - Or add one during API action “start-subscription” in parameter manage_subscription_url_callback

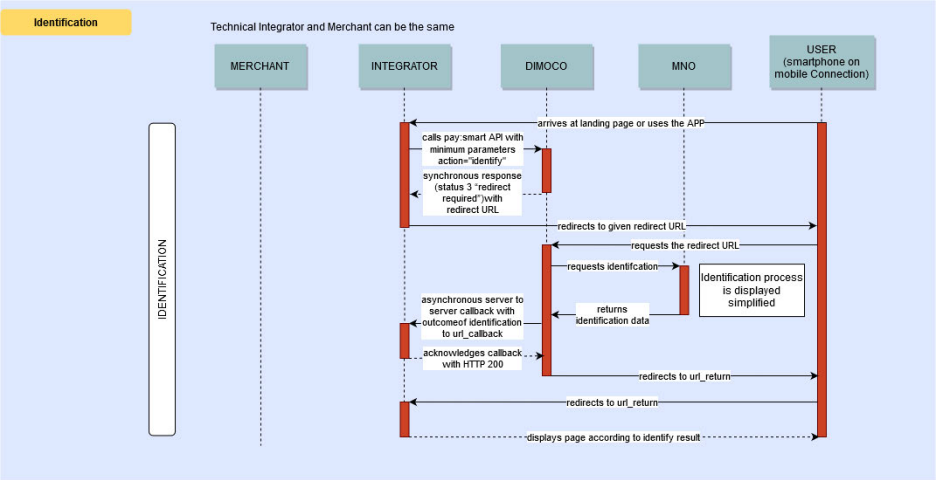
6. Static message texts (handled by DIMOCO)

- FAGG MT: "Vielen Dank für die Anmeldung zu {service_name} für {amout} EUR/{frequency}. Rechtliche Infos: {contract_confirmation_url} - Zugang zum Dienst: {content_access_url} Hilfe: {hotline}"
- Stop Confirmation MT: "Der Dienst von {merchant_name} - {service_name} wurde beendet."

7. Identification & Validation

7.1. Identification:

1. USER arrives at INTEGRATOR hosted landing page or uses the APP of the INTEGRATOR
2. INTEGRATOR calls DIMOCO pay:smart API with minimum parameters
    - action (=”identify”)
    - merchant, order, digest (These are credentials provided by DIMOCO; digest is calculated by INTEGRATOR using a provided shared secret)
    - request_id (unique INTEGRATOR created ID)
    - url_callback (URL where DIMOCO will send the asynchronous callback)
    - url_return (URL where DIMOCO returns the USER after the identification)
    - service_name
3. DIMOCO returns synchronous response to INTEGRATOR (status 3 “redirect required”) alongside with a redirect URL
4. INTEGRATOR redirects USER to given redirect URL
    - This redirect does not need to be visible to the USER, no interaction of the USER is needed
5. DIMOCO tries to identify USER
6. DIMOCO sends asynchronous server to server callback with the outcome of the identification to the INTEGRATOR's url_callback
7. INTEGRATOR acknowledges callback with HTTP 200
8. DIMOCO redirects the USER to the given url_return
9. Based on the outcome of the identify
    - In case channel = wap the USER is in mobile web -> continue with “**CLICK Subscription Flow**”
    - In case channel = web the USER is in Wifi/on Desktop –> continue with “**WEB TAN Subscription Flow**”



7.2. Operator Lookup (asynch):

1. INTEGRATOR displays MSISDN entry page to the USER
2. USER enters MSISDN
3. INTEGRATOR calls DIMOCO pay:smart API with minimum parameters
    - action (="operator-lookup")
    - merchant, order, digest (This are Credentials provided by DIMOCO, Digest is calculated by INTEGRATOR using a provided shared secret)
    - request_id (unique INTEGRATOR created ID)
    - url_callback (URL where DIMOCO will send the asynchronous Callback)
    - service_name
    - msisdn = USER's MSISDN
4. DIMOCO returns synchronous response to INTEGRATOR (status 5 "pending")
5. DIMOCO identifies USER's MNO
6. DIMOCO sends a server to server callback with the outcome of the lookup INTEGRATORs url_callback
    - Action Result Status = 0 and operator = valid operaor -> operator detected
    - Action Result Status = 1 -> operator detection failed
    - Full Callback Explanation can be found in the pay:smart API Specification Document
7. INTEGRATOR acknowledges callback with HTTP 200

7.3. Operator Lookup (synch):

1. INTEGRATOR displays MSISDN entry page to the USER
2. USER enters MSISDN
3. INTEGRATOR calls DIMOCO pay:smart API with minimum parameters
    - action (="operator-lookup")
    - merchant, order, digest (This are Credentials provided by DIMOCO, Digest is calculated by INTEGRATOR using a provided shared secret)
    - request_id (unique INTEGRATOR created ID)

    - url_callback (URL where DIMOCO will send the asynchronous Callback)
    - service_name
    - msisdn = MSISDN entered by USER
4. DIMOCO identifies USER's MNO
5. DIMOCO returns synchronous response to INTEGRATOR
    - Action Result Status = 0 and operator = valid operaor -> operator detected
    - Action Result Status = 1 -> operator detection failed
    - Full Callback Explanation can be found in the pay:smart API Specification Document
6. INTEGRATOR acknowledges callback with HTTP 200

8. Subscription Flow

8.1. CLICK Subscription Flow (Pages by MNO):

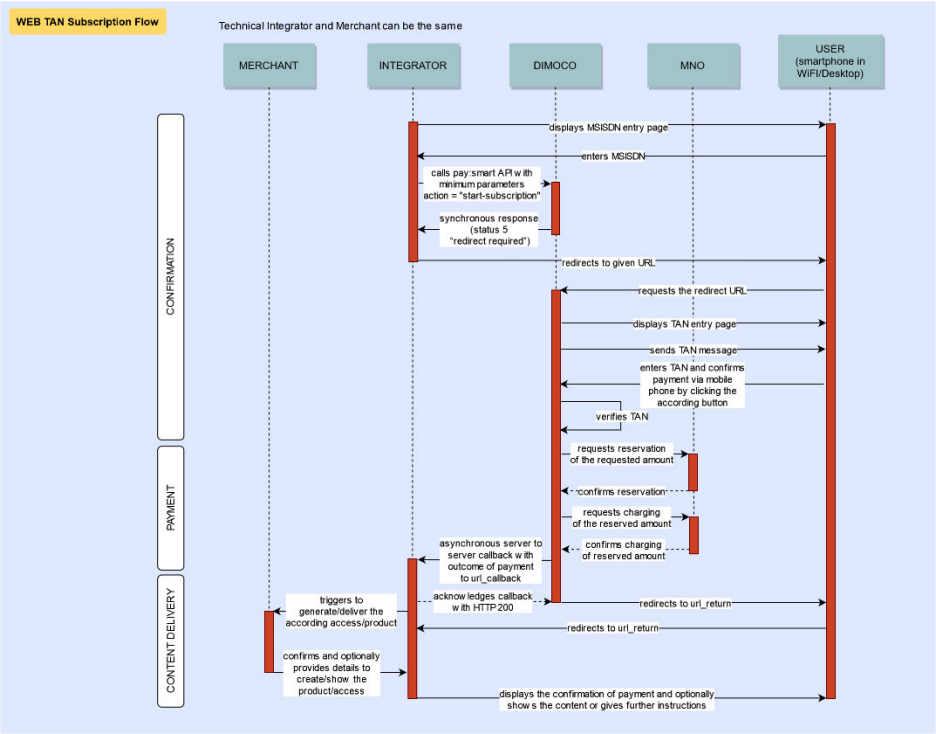
1. INTEGRATOR displays landing page to the USER
    1. INTEGRATOR processes **identify** or **operator-lookup**
2. INTEGRATOR calls DIMOCO pay:smart API with minimum parameters
    - action (=”start-subscription”)
    - merchant, order, digest (These are credentials provided by DIMOCO; digest is calculated by INTEGRATOR using a provided shared secret)
    - request_id (unique INTEGRATOR created ID)
    - amount
    - url_callback (URL where DIMOCO will send the asynchronous callback)
    - url_return (URL where DIMOCO will redirect the USER after confirmation)
    - service_name
    - transaction = TransactionID from the previous Identify
    - (optional) prompt_merchant_args = {"logo":{"img":"https://source.path.to/company/logo.png","alt":"Company Name"}}
    - (optional) prompt_product_args = {"pic":{"img":"https://source.path.to/product/image.png","alt":"Product Name"},"desc":{"de":"Prodcut description","en":"Prodcut description"}}
    - Other parameters may be necessary to improve USER experience (see also pay:smart API specification document)
3. DIMOCO returns synchronous response to INTEGRATOR (status 3 “redirect required”) alongside with a redirect url
4. INTEGRATOR redirects USER to given redirect url
5. USER confirms subscription via mobile phone by clicking the according button on MNO's payment page
6. DIMOCO tries to bill the requested amount at the USER's MNO
7. DIMOCO sends asynchronous server to server callback with the outcome of the payment and subscription to the INTEGRATOR's url_callback
    - In case action_result status = 0 and billed_amount = amount -> payment and subscription successful

    - In case action_result status = 1 and billed_amount = 0 (alongside with error details) -> payment and subscription failed
    - Full callback explanation can be found in the pay:smart API specification document
8. INTEGRATOR acknowledges callback with HTTP 200
9. DIMOCO redirects the USER to the given url_return
10. (INTEGRATOR displays the outcome of the subscription to the USER)
11. INTEGRATOR displays the content/redirects to MERCHANT's portal/gives further instructions to USER

8.2. WEB TAN Subscription Flow (Pages and TAN by MNO):

1. INTEGRATOR displays MSISDN entry page to the USER
2. USER enters MSISDN
    - INTEGRATOR processes **identify** or **operator-lookup**
3. INTEGRATOR calls DIMOCO pay:smart API with minimum parameters
    - action (=”start-subscription”)
    - merchant, order, digest (These are Credentials provided by DIMOCO; digest is calculated by INTEGRATOR using a provided shared secret)
    - request_id (unique INTEGRATOR created ID)
    - amount
    - url_callback (URL where DIMOCO will send the asynchronous callback)
    - service_name
    - msisdn = USER's MSISDN
    - operator = USER's MNO
    - (optional) prompt_merchant_args = {"logo":{"img":"https://source.path.to/company/logo.png","alt":"Company Name"}}
    - (optional) prompt_product_args = {"pic":{"img":"https://source.path.to/product/image.png","alt":"Product Name"},"desc":{"de":"Prodcut description","en":"Prodcut description"}}
    - Other parameters may be necessary to improve USER experience (see also pay:smart API specification document)
4. DIMOCO returns synchronous response to INTEGRATOR (status 3 “redirect required”) alongside with a redirect URL
5. INTEGRATOR redirects USER to given redirect URL
6. MNO sends out the TAN message to the USER
7. USER enters the TAN on the TAN entry page and confirms subscription via mobile phone by clicking the according button on MNO's payment page
8. DIMOCO tries to bill the requested amount at the USER's MNO
9. DIMOCO sends asynchronous server to server callback with the outcome of the payment and subscription to the INTEGRATOR's url_callback
    - In case action_result status = 0 and billed_amount = amount -> payment and subscription successful

    - In case action_result status = 1 and billed_amount = 0 (alongside with error details) -> payment and subscription failed
    - Full callback explanation can be found in the pay:smart API specification document
10. INTEGRATOR acknowledges callback with HTTP 200
11. DIMOCO redirects the USER to the given url_return
12. (INTEGRATOR displays the outcome of the subscription to the USER)
13. INTEGRATOR displays the content/redirects to MERCHANT's portal/gives further instructions to USER



9. Renewal Flow

9.1. Renewals - handled by DIMOCO:

1. On schedule DIMOCO tries to bill the requested amount at the USER's MNO
2. DIMOCO sends asynchronous server to server callback with the outcome of the payment and corresponding subscriptionID to the INTEGRATOR's manage_subscription_url_callback
    - In case action_result status = 0 and billed_amount = amount -> payment successful
    - In case action_result status = 1 and billed_amount = 0 (alongside with error details) -> payment failed; may be re-scheduled according to retry policy
    - Full callback explanation can be found in the pay:smart API specification document

3. INTEGRATOR acknowledges callback with HTTP 200
4. INTEGRATOR confirms to MERCHANT
5. MERCHANT prolongs USER's access

10. Unsubscription Flow

10.1. Unsubscribe - If USER selects to unsubscribe by clicking on the corresponding button on the merchant's page:

1. USER confirms unsubscription by clicking the according button on MERCHANT’s portal
2. INTEGRATOR calls DIMOCO pay:smart API with minimum parameters
    - action (="close-subscription")
    - merchant, order, digest (These are Credentials provided by DIMOCO; digest is calculated by INTEGRATOR using a provided shared secret)
    - request_id (unique INTEGRATOR created ID)
    - url_callback (URL where DIMOCO will send the asynchronous callback)
    - service_name
    - subscription = corresponding subscriptionID (retrieved from "start-subscription" callback)
3. DIMOCO unsubscribes the USER
4. DIMOCO sends Stop confirmation message to the USER
5. DIMOCO sends asynchronous server to server callback with the outcome of the unsubscription and corresponding subscriptionID to the INTEGRATOR's url_callback
6. INTEGRATOR acknowledges callback with HTTP 200
7. After expiry of the already pre-paid period (grace period) MERCHANT blocks the USER's access to content

10.2. Unsubscribe - If USER selects to unsubscribe by clicking on the corresponding button on DIMOCO’s/MNO’s customer care portal page:

1. USER confirms unsubscription by clicking the according button on DIMOCO's/MNO's portal
2. DIMOCO unsubscribes the USER
3. DIMOCO sends Stop confirmation message to the USER
4. DIMOCO sends asynchronous server to server callback with the outcome of the unsubscription and corresponding subscriptionID to the INTEGRATOR's close_notification_url_callback
5. INTEGRATOR acknowledges callback with HTTP 200

6. After expiry of the already pre-paid period (grace period) MERCHANT blocks the USER's access to content

11. Refund Flow:

1. INTEGRATOR calls DIMOCO pay:smart API with minimum parameters
    - action (=”refund”)
    - merchant, order, digest (These are credentials provided by DIMOCO; digest is calculated by INTEGRATOR using a provided shared secret)
    - request_id (unique INTEGRATOR created ID)
    - url_callback (URL where DIMOCO will send the asynchronous callback)
    - service_name
    - transaction = TransactionID from referred payment
2. DIMOCO returns synchronous response to INTEGRATOR (status 3 “pending”)
3. DIMOCO tries to refund the requested amount at the USER's MNO
4. DIMOCO sends asynchronous server to server callback with the outcome of the refund to the INTEGRATOR's url_callback
    - In case action_result status = 0 and billed_amount = amount -> refund successful
    - In case action_result status = 1 and billed_amount = 0 (alongside with error details) -> refund failed
5. Full callback explanation can be found in the pay:smart API specification document
6. INTEGRATOR acknowledges callback with HTTP 200
7. INTEGRATOR confirms refund to MERCHANT
8. INTEGRATOR confirms to MERCHANT and optionally provides necessary details to create/show the product/access on USER's device

12. Additional features:

12.1. Free SMS MT sending:

1. INTEGRATOR calls DIMOCO pay:smart API with minimum parameters
    - action (=”prompt”)
    - merchant, order, digest (This are Credentials provided by DIMOCO, Digest is calculated by INTEGRATOR using a provided shared secret)
    - request_id (unique INTEGRATOR created ID)
    - url_callback (URL where DIMOCO will send the asynchronous Callback)
    - service_name
    - msisdn = USER's MSISDN
    - operator = USER's MNO
    - subject = content
    - channel = sms
    - prompt_content_args = {"text":{"de":"Free MT text""}}
2. DIMOCO returns synchronous response to INTEGRATOR (status 5 “pending”)
3. DIMOCO sends out the free MT message to the USER

4. DIMOCO sends a server to server callback with the outcome of the delivery of the TAN message to the INTEGRATORs url_callback
    - Action Result Status = 0 -> Free MT message delivered
    - Action Result Status = 1 -> Free MT message delivery failed
    - Full Callback Explanation can be found in the pay:smart API Specification Document
5. INTEGRATOR acknowledges callback with HTTP 200

12.2. Blocklist

1. INTEGRATOR calls DIMOCO pay:smart API with minimum parameters
    1. action (="add-blocklist"/"remove-blocklist"/"check-blocklist")
    2. merchant, order, digest (This are Credentials provided by DIMOCO, Digest is calculated by INTEGRATOR using a provided shared secret)
    3. request_id (unique INTEGRATOR created ID)
    4. url_callback (URL where DIMOCO will send the asynchronous Callback)
    5. msisdn = USER's MSISDN
    6. operator = USER's MNO
    7. blocklist_scope (merchant or order; in case of "check-blocklist" not mandatory)
2. DIMOCO returns synchronous response to INTEGRATOR (status 5 “pending”)
3. DIMOCO sets USER's msisdn on requested blocklist level (merchant/order)
4. DIMOCO sends a server to server callback with the outcome of the blocklist to the INTEGRATORs url_callback
    1. Action Result Status = 0 -> blocklist action successful
    2. Action Result Status = 1 -> blocklist action failed
    3. Full callback explanation can be found in the pay:smart API specification document
5. INTEGRATOR acknowledges callback with HTTP 200